

ShieldIT FSX:

Compliance-Driven Secure Communications & Mobile Threat Defense for Financial Services, Federal Deployments, and Highly Regulated Environments

Executive Summary

Financial institutions, capital markets, federal agencies, and other highly regulated sectors operate under some of the world's strictest cybersecurity, communication retention, and supervisory governance requirements. As mobile-first operations expand across trading, advisory, executive, and mission-critical environments, organizations must protect sensitive communications end-to-end while ensuring full alignment with regulatory expectations for auditability, retention, and mobile security posture.

ShieldIT FSX by ASPIS Cyber is a **compliance-driven secure communications and mobile threat defense platform** engineered specifically for **financial services, federal deployments, and high-assurance regulated environments**. FSX integrates advanced encrypted communications, policy-driven recording, mobile and endpoint threat defense, and compliance-aligned governance into a **single, unified zero-trust platform** capable of operating in the most demanding oversight and security scenarios.

ShieldIT FSX delivers



End-to-end encrypted communications

chat, voice, video, file sharing



Encrypted call & video recording

with policy-based enablement and controlled retention



SRTP-over-TLS media protection

for high-assurance confidentiality and integrity



Advanced Mobile Threat Defense (MTD)

and endpoint risk detection across iOS, Android, Windows, and macOS



Compliance-aligned audit logging, retention workflows and supervisory visibility



Deployment flexibility

across Cloud, Private Cloud, Hosted, Sovereign Cloud, On-Premise, and fully isolated environments

Engineered for banks, broker-dealers, trading desks, wealth managers, treasury operations, federal agencies, and other compliance-bound entities, ShieldIT FSX aligns with the supervisory principles and expectations of **SEC 17a-4, FINRA, GLBA, PCI DSS, SOX, FFIEC**, and federal frameworks including **FISMA, NIST 800-53, and CMMC**.

With FSX, organizations gain a unified, audit-ready platform that ensures **secure, compliant, and resilient communications** across hybrid, BYOD, distributed, and mission-critical operational environments—without compromising performance, usability, or security.

Key Use Cases for Financial, Federal, and Highly Regulated Environments

Regulated Communications Across Trading, Advisory & Federal Operations

- **Encrypted, recorded voice and video communications** for trading desks, supervisory groups, mission operations, and federal program offices.
- **Federation and communication boundary controls** aligned to internal policy, conduct standards, and federal/financial communication governance requirements.

Client Advisory, M&A, Sensitive Transactions & Interagency Collaboration

- **Secure, compliant collaboration** through encrypted messaging, voice, video, and file sharing across distributed deal teams, advisors, and interagency partners.
- **Policy-based retention and recording** for high-value client interactions, confidential transactions, due diligence, and sensitive federal coordination.

Compliance, Supervisory Oversight & Federal Audit Alignment

- **Immutable communication logs**, recording retention, and automated monitoring supporting SEC/FINRA, SOX, GLBA, FFIEC, FISMA, and NIST-aligned supervisory workflows.
- **AuditBot-driven visibility** into privileged actions, administrative changes, and communications touching regulated or mission-critical environments.

Hybrid Workforce, BYOD Security & Cross-Agency Mobility

- **Zero-trust enforcement** across traders, bankers, advisors, executives, federal personnel, contractors, and mission support teams.
- **Continuous device posture monitoring** for personal, corporate, and contractor devices — ensuring only secure, compliant endpoints access sensitive systems.

Data Residency, Sovereignty & Cross-Border / Federal Mandates

- Deployable in **Cloud, Private Cloud, Hosted, Sovereign Cloud, On-Premises, or Air-Gapped** environments for maximum regulatory alignment.
- Supports **national data sovereignty, cross-border financial operations, and federal mission requirements**, ensuring compliance with jurisdiction-specific retention and localization laws.

High-Assurance, Mission-Critical Communications

- Enables **secure, policy-controlled communication** for roles requiring heightened confidentiality — such as executive leadership, trading supervisors, federal program managers, and high-risk operational units.
- Provides **hardened communication pathways** suitable for environments with elevated threat profiles, insider-risk exposure, or classified dependencies.

To learn more or to arrange a consultation, visit www.aspiscyber.com
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Core Features of ShieldIT FSX

Advanced Secure Communications & Encrypted Recording

- ✓ End-to-end encrypted **chat, voice, video, and file sharing** using AES-256, Olm/Megolm, and Double-Ratchet cryptographic protocols.
- ✓ **Secure, policy-driven call and video recording** with encryption in transit (SRTP/TLS) and at rest for regulated and mission-critical communications.
- ✓ **SRTP-over-TLS** protection for real-time media streams, preventing interception or replay across mobile, desktop, and cross-network environments.
- ✓ **Role-based recording controls** (e.g., record trading desks; exempt executives, legal, or privileged roles).
- ✓ **Retention policies** aligned with SEC 17a-4, FINRA, SOX, GLBA, PCI DSS, FFIEC, FISMA, and NIST supervision expectations.

AuditBot, Communications Capture & Compliance Logging

- ✓ **AuditBot-based privileged action logging**, tracking administrative changes, access events, and configuration modifications.
- ✓ **Automatic capture of written communications, shared files, and system events** for regulatory and federal supervision.
- ✓ **Immutable, cryptographically signed audit logs** supporting evidentiary integrity, eDiscovery, and litigation hold.
- ✓ **Compliance-ready archival & export** to Microsoft Purview, Smarsh, Global Relay, or customer-designated systems.
- ✓ **Off-channel communication detection for** identifying unauthorized or non-compliant messaging pathways.

Mobile Threat Defense (MTD) & Device Integrity Enforcement

- ✓ On-device AI detection of **phishing, malware, SIM swaps, rogue apps, and behavioral anomalies** across iOS and Android.
- ✓ **Rogue Wi-Fi / MITM detection and auto-blocking** across mobile and laptop endpoints.
- ✓ **Root/jailbreak detection and OS integrity monitoring** with automated remediation.
- ✓ **Device quarantine and conditional access** for high-risk or non-compliant endpoints.

Unified Endpoint Security (Mobile + Desktop)

- ✓ **Embedded on-device MTD** for mobile endpoints.
- ✓ **Desktop Security controls** for Windows and macOS devices, including configuration integrity, harmful application detection, network threat detection, and posture verification.
- ✓ Behavioral risk monitoring to identify suspicious user actions and potential insider threats.
- ✓ **Zero-trust enforcement** across all device types—mobile, desktop, BYOD, remote, and contractor systems.

Identity, Access & Federation Controls

- ✓ Native support for **Azure AD / Entra ID, Active Directory, Okta, Ping Identity, SAML/OIDC, and SCIM.**
- ✓ **Role-based access control (RBAC)** mapped to financial, federal, and regulated roles (Trader, Advisor, Executive, Compliance Officer, Federal Program Manager, Contractor).

- ✓ **Federation policies** defining:
 - Who may communicate internally.
 - Which roles may communicate externally (e.g., advisors ↔ clients, regulators ↔ compliance).
 - Which roles are restricted entirely.
- ✓ **Cross-domain federation** for multi-bank groups, subsidiaries, and federal interagency collaboration.
- ✓ **Automated provisioning/deprovisioning** via directory synchronization.

Microsoft Intune & MDM/EMM Enforcement

- ✓ **Embedded Intune posture enforcement**, ensuring compliance before communication access is granted.
- ✓ Support for **VMware Workspace ONE, MobileIron, and other EMM/MDM platforms** for environment-wide enforcement.
- ✓ **Conditional access** triggered by device risk, OS integrity, user behavior, or geo-location.
- ✓ **Automated remediation** for compromised or noncompliant devices.

Data Loss Prevention (DLP) & Zero-Trust Controls

- ✓ Restrictions on **copy/paste, screenshots, and external file sharing** across regulated channels.
- ✓ **Geo-fencing** aligned to trading floors, advisory zones, federal sites, and mission-specific boundaries.
- ✓ **Workspace isolation** separating personal and enterprise data (BYOD-safe).
- ✓ **Zero-trust validation** at every authentication, messaging, and session event.

Deployment Flexibility & Sovereign/Federal Readiness

- ✓ Deployable in **Cloud, Private Cloud, Hosted, Sovereign Cloud, On-Premise, or Air-Gapped** architectures.
- ✓ **Dedicated, isolated clusters** for high-assurance banking and federal security requirements.
- ✓ **Zero-knowledge encryption** and **customer-controlled keys** for complete data sovereignty.
- ✓ Alignment with **FISMA, NIST 800-53, CMMC**, and financial-sector sovereignty requirements across regions.

Centralized Governance with ManageIT

- ✓ **Unified dashboard** for communications oversight, recording visibility, device posture, risk alerts, and retention workflows.
- ✓ **Real-time alerting** for high-risk events, compliance deviation, and potential audit exceptions.
- ✓ **Cross-tenant management** for large institutions, multi-entity groups, and federal division structures.
- ✓ **SIEM/XDR forwarding** to Microsoft Sentinel, Splunk, QRadar, etc. for advanced threat correlation.

High-Assurance, Mission-Critical Communications

- ✓ Hardened communication pathways for **trading supervisors, federal program officers, executive leadership, intelligence-adjacent roles**, and high-risk mission units.
- ✓ Consistent, secure communication across **remote, hybrid, BYOD, operational, and classified-aligned environments**.

Key Differentiators of ShieldIT FSX

01. Compliance-Driven Secure Communications for Regulated Environments

ShieldIT FSX is engineered specifically for **financial, federal, and highly regulated sectors**, delivering encrypted communications, call/video recording, and supervisory governance aligned to **SEC, FINRA, GLBA, SOX, FFIEC, FISMA, NIST 800-53, and CMMC** frameworks.

Unlike consumer apps or standalone recording tools, FSX provides **policy-based, enforceable, and audit-ready** secure communication across all channels.

02. SRTP-over-TLS Media Security for Voice & Video

FSX combines **SRTP media encryption** with **TLS-secured signaling**, providing high-assurance protection for voice/video communications across mobile networks, Wi-Fi, private infrastructure, or cross-border environments.

This security level is typically available only in **military-grade or federal-grade** systems — now delivered in a financial-compliant platform.

03. Integrated Call & Video Recording with Regulatory Retention Controls

FSX offers **built-in, encrypted call and video recording** with retention mapped to supervisory mandates (SEC 17a-4, FINRA, federal audit retention).

Recordings are secured, indexed, time-stamped, and exportable for compliance review or eDiscovery — without third-party integration or fragmented workflows.

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04. Unified Security Coverage for Mobile and Desktop Endpoints

FSX combines **on-device Mobile Threat Defense (MTD)** with **desktop security**, providing global, cross-platform enforcement for iOS, Android, Windows, and macOS.

This unified model eliminates multi-vendor operational complexity and creates a consistent **zero-trust enforcement layer** across all endpoints.

05. Zero-Trust Enforcement Across All Communication Channels

Every communication session — chat, voice, video, and file exchange — undergoes:

- Device posture validation.
- Identity verification.
- Policy compliance checks.
- Conditional access gating.

This extends zero-trust to **all communication pathways**, not just network access.

06. Federation Controls for Internal, External & Interagency Boundaries

FSX enables institutions to define exactly who can communicate with whom, including:

- Internal role separation (Trader ↔ Compliance; Advisor ↔ Client; Federal Officer ↔ External Unit).
- Cross-entity restrictions (Subsidiaries, affiliates, or partner firms).
- Interagency boundaries for federal deployments.

This is critical for preventing **"off-channel" violations** and ensuring supervisory alignment.

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07. Fully Flexible Deployment Models (Cloud → Air-Gapped)

FSX deploys in:

- Cloud
- Private Cloud
- Hosted
- Sovereign Cloud
- On-Premises
- Air-Gapped environments

This flexibility supports **national data sovereignty**, federal requirements, and cross-border regulatory obligations.

08. Customer-Controlled Keys & Zero-Knowledge Architecture

Organizations maintain full ownership of encryption keys and retention storage. FSX operates on a **zero-knowledge** model — ASPIS cannot access customer data, communications, recordings, or stored retention archives.

09. Integrated Compliance Governance (AuditBot + ManageIT)

FSX integrates **AuditBot** and **ManageIT** to provide:

- Privileged action visibility
- Compliance and supervisory workflows
- Centralized retention oversight
- Real-time risk alerts
- Unified communications governance

This creates a **single pane of glass** for compliance, IT security, and supervisory teams.

10. High-Assurance Operation for Mission-Critical Workflows

FSX supports executive, trading, federal, intelligence-adjacent, and high-risk operational units, ensuring secure communications in environments requiring:

- Low-latency encrypted voice/video
- BYOD support under strict policy
- Multi-region redundancy
- Federal compliance alignment
- Zero downtime expectations

Architecture Overview

ShieldIT FSX is built on a **cloud-native, microservices-based architecture** engineered for the stringent requirements of financial institutions, federal environments, and highly regulated operational domains. The platform integrates dedicated services for **encrypted media capture, SRTP/TLS negotiation, compliance retention, audit integrity, mobile threat defense, and desktop security**, all within a unified zero-trust framework.

Core Architectural Components

Kubernetes-Orchestrated Microservices for Elasticity & High Availability

- Each FSX capability—communications, recording, compliance logging, threat detection, identity enforcement—is delivered as an independent microservice.
- Kubernetes ensures **auto-scaling, failover, rolling updates, and global resiliency** across regulated workloads.

Dedicated Media Recording & Encryption Nodes

- Secure call and video recording handled by **isolated media processing nodes** with SRTP/TLS negotiation.
- Real-time media flows are encrypted end-to-end and processed in dedicated clusters to maintain **retention integrity** and meet regulated recording requirements.

Encrypted Storage Clusters for Recorded Communications

- Voice, video, chat transcripts, and file exchanges are stored in **AES-256 encrypted storage backends** with optional WORM (Write Once, Read Many) retention.
- Supports **data sovereignty, cross-border controls, and financial/federal archival mandates**.

Immutable Audit Log Pipelines

- All privileged actions, communication events, policy updates, device posture changes, and recording metadata flow into **tamper-resistant audit pipelines**.
- Logs are cryptographically signed, time-stamped, and stored in compliant audit backends supporting supervisory review and evidentiary standards.

API Connectors for Compliance Archival

- Native integration with leading compliance and regulatory platforms, including:
 - **Microsoft Purview**
 - **Smarsh**
 - **Global Relay**
 - Customer-hosted or sovereign archival systems
- Enables seamless routing of recordings, transcripts, and audit logs into regulated retention systems.

Zero-Knowledge Encryption with Customer-Controlled Certificates

- FSX enforces a **zero-knowledge security model**: ASPIS cannot access data, recordings, audit logs, or message content.
- Organizations maintain full ownership of encryption keys, with support for HSM-backed and sovereign key management.

Modular Architecture Enables:



Real-Time Threat Detection

On-device Mobile Threat Defense (MTD) and Desktop Security modules feed risk telemetry into backend correlators for immediate action.



Encrypted Media Recording & Regulatory Retention

Voice and video are encrypted, captured, retained, and exported per-regulatory timelines.



Multi-Region & Sovereign Deployments

Supports financial institutions operating across the U.S., EU, APAC, Middle East, and federal/sovereign clouds.



Full Tenant Isolation & Compliance Segmentation

Each institution, subsidiary, business unit, or federal entity runs in isolated namespaces with strict RBAC and cryptographic boundaries.

Licensing & Delivery Model

ShieldIT FSX offers flexible licensing and deployment options designed to meet the operational, security, and regulatory requirements of financial institutions, federal agencies, and highly regulated organizations. The model ensures predictable cost structures, scalable growth, and alignment with supervisory retention and sovereignty mandates.

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Per-User Annual Licensing

- Straightforward **per-user, annual subscription model** with tiered pricing for large institutions, multi-entity groups, and federal environments.
- Volume-based discounts available for enterprise-scale deployments, global financial institutions, and federal programs.
- Optional add-ons for specialized compliance or sovereign deployment requirements.

High-Security Deployment Options

- FSX can be deployed in **Cloud, Private Cloud, Hosted, Sovereign Cloud, On-Premise, or Air-Gapped environments**, supporting financial and federal sovereignty mandates.
- Infrastructure can be provisioned as:
 - **Dedicated isolated clusters**
 - **Institution-controlled instances**
 - **Sovereign or region-specific nodes** for data residency compliance
- Supports **national data residency, cross-border controls**, and **federal enclave deployments**.

Zero-Knowledge Encryption & Customer Key Ownership

- Customers retain full control over encryption keys, supporting HSM-backed and sovereign key management policies.
- Zero-knowledge architecture ensures ASPIS Cyber has **no access** to communications, recordings, retention archives, or audit logs.

Service Level Commitments

- ✓ SLA-backed performance, uptime guarantees, and response time commitments tailored to financial and federal operational environments.
- ✓ High-availability and multi-zone redundancy options for trading desks, wealth divisions, federal program operations, and mission-critical teams.
- ✓ 24/7 global support for critical issues impacting compliance, communications, or mobile security posture.

Implementation & Compliance Onboarding

- Guided deployment with ASPIS engineering, ensuring rapid adoption and proper configuration of:
 - Communication retention policies
 - Recording controls
 - Mobile and desktop security enforcement
 - Identity & federation mapping
 - Compliance archival workflows
- Optional **Compliance Architecture Workshops** for internal audit, supervisory, and governance teams.

Optional Add-On Services

- **ShieldIT Black** hardened devices for high-risk roles such as trading supervisors, federal program officers, or executive leadership.
- **PSTN voice services** with encrypted signaling and recording pathways.
- **Extended retention packages** for long-term compliance storage.

Conclusion

In an era where financial institutions, federal agencies, and regulated enterprises face unprecedented oversight, cyber risk, and operational complexity, the need for **secure, compliant, and tightly governed communications** has never been greater. ShieldIT FSX delivers a comprehensive, zero-trust platform designed to safeguard sensitive workflows, protect high-value communications, and align with the strict supervisory requirements that define financial and federal environments.

By combining **end-to-end encrypted communications, policy-driven call and video recording, SRTP-over-TLS media protection, mobile and desktop security, and compliance-aligned retention and audit governance**, ShieldIT FSX offers a level of visibility, control, and operational resilience unmatched by consumer tools or fragmented enterprise solutions.

FSX enables organizations to:

- Protect critical advisory, trading, supervisory, and mission communications
- Maintain continuous compliance alignment with financial and federal standards
- Enforce zero-trust device posture across hybrid, BYOD, and distributed workforces
- Strengthen operational resilience through sovereign, on-premise, or air-gapped deployments
- Support supervisory review, audit readiness, and regulatory reporting with immutable, encrypted data retention

By adopting ShieldIT FSX, institutions gain a unified, future-ready foundation for **secure, compliant, and mission-critical communication**, enabling them to operate with confidence in the most demanding regulatory, financial, and federal environments.

ShieldIT FSX is engineered for organizations that cannot afford compromise, delivering the security, governance, and assurance required to protect the communications that power global finance, national-level missions, and high-trust regulated operations.

For more information or to discuss deployment options, contact:
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ASPIS Cyber Technologies